Gabriele Mauro

Torino, Italy
gabriele.mauro2004@gmail.com linkedin.com/in/gabriele-mauro2004 github.com/Gale4409

Education

Politecnico di Torino **Sep 2024 – Jul 2027 (expected)**
Bachelor of Science in Computer Engineering *Torino, Italy*

- GPA: 28.74/30 - Major GPA: 29.48/30
- **Relevant coursework:** Computer Science, Algorithms & Data Structures, Linear Algebra & Geometry, Calculus I & II, Electromagnetics & Circuit Theory, Physics I, Computer Systems

Research Interests

Algorithm Design & Optimization, Graph Theory & Network Analysis, Dynamic Data Structures, Combinatorial Optimization, Computational Complexity

Publications

[S1] G. Goranci[1], M. Probst Gutenberg[2], P. Kiss[1], **G. Mauro**[1], G. Zöcklein[2], "Dynamic Spectral Sparsification: Theory and Practice", *under review at ICLR 2027.*
[1]University of Vienna [2]ETH Zurich *(authors in alphabetical order)*

Experience

Undergraduate Student Researcher **Oct 2026 – Present**
Politecnico di Torino (Supervisor: Prof. Stefano Quer) *Torino, Italy*

- Working toward a practical implementation of recent all-pairs max-flow (Gomory–Hu tree) algorithms.

Visiting Researcher **Mar 2026 – Oct 2026**
University of Vienna (Supervisor: Prof. Gramoz Goranci) *Vienna, Austria (Hybrid)*

- Engineered, to our knowledge, the first practical implementation of the fully dynamic spectral graph sparsifier proposed by Abraham et al. (FOCS 2016).
- Co-designed a new dynamic sparsifier based on decremental low-diameter decompositions, which outperforms the Abraham et al. baseline in practice [S1].

ChimpCop **2020 – 2022**
Co-Founder & Operations Manager *Remote*

- Co-founded an online e-commerce automation community that grew to 1,500+ members; membership fees covered running costs only.
- Ran the technical side: CLI scripts, proxy networks and automation tools for high-traffic product releases.

Teaching

Teaching Assistant – Algorithms & Data Structures | *Politecnico di Torino* **Sep 2026 – Present**

Tutor – Physics I | *Politecnico di Torino* **Mar 2026 – Jul 2026**

- Selected for an official university grant to provide 1-on-1 mentorship in kinematics, dynamics, and thermodynamics to students with Specific Learning Disorders (DSA) and Special Educational Needs (BES).

Projects

DynaGraph: fully dynamic graph connectivity library in C Feb – Mar 2026
TSP Optimizer: Branch & Bound solver in C with custom hash tables Feb 2026
Social Network Graph Analyzer: SCC detection (Tarjan) and triadic closure prediction in C Feb 2026

Activities

Mentee – Lead The Future (LTF) | *Italian STEM mentorship programme, 2027 cohort* **Sep 2026 – Present**

Technical Skills

Programming Languages: C, C++, Python, RISC-V
Tools & Technologies: Git, GitHub, CMake, Valgrind
Areas of Expertise: Algorithm Design & Analysis, Dynamic Data Structures, Graph Theory, Complexity Analysis, Memory Management, Software Engineering Principles
Spoken Languages: Italian (Native), English (C1, IELTS 8.0)